Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R12-E1_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)

Name	SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)
Description	
Analysis	Stress, strain/ simplified loading

Design

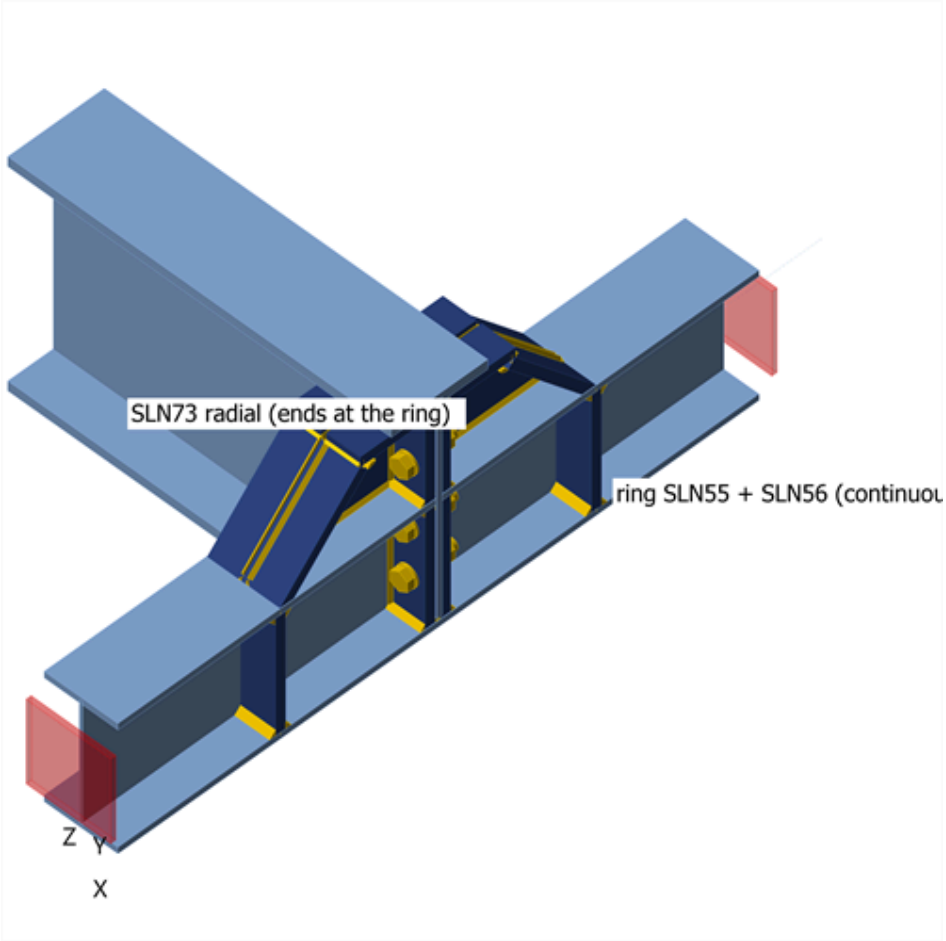
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73 radial (ends at the ring)	1 - UB457x191x98	0.0	0.0	0.0	76	0	0
ring SLN55 + SLN56 (continuous)	2 - UB254x146x43	-90.0	0.0	0.0	0	0	-104

Supports and forces

Name	Support	Forces in	X [mm]
SLN73 radial (ends at the ring) / begin		Position	0
ring SLN55 + SLN56 (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
ring SLN55 + SLN56 (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0



SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous), Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M24 10.9	24	900.0	1000.0	452

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_102	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_103	SLN73 radial (ends at the ring) / Begin	9.6	-0.1	-86.2	-0.1	-45.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_104	SLN73 radial (ends at the ring) / Begin	7.9	0.0	-86.4	-0.1	-45.4	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_105	SLN73 radial (ends at the ring) / Begin	8.8	-0.1	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_106	SLN73 radial (ends at the ring) / Begin	8.0	0.0	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_107	SLN73 radial (ends at the ring) / Begin	1.8	-0.1	-7.8	0.0	-4.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_108	SLN73 radial (ends at the ring) / Begin	1.7	0.1	-8.4	0.0	-4.6	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_109	SLN73 radial (ends at the ring) / Begin	1.4	0.1	-8.1	0.0	-4.5	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_110	SLN73 radial (ends at the ring) / Begin	0.6	-0.1	-8.1	0.0	-4.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_111	SLN73 radial (ends at the ring) / Begin	7.4	-0.1	-61.7	0.0	-32.6	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_112	SLN73 radial (ends at the ring) / Begin	6.3	0.1	-62.3	0.0	-32.8	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_113	SLN73 radial (ends at the ring) / Begin	6.7	0.0	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_114	SLN73 radial (ends at the ring) / Begin	5.6	-0.1	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_115	SLN73 radial (ends at the ring) / Begin	2.4	-0.1	-13.1	0.0	-7.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_116	SLN73 radial (ends at the ring) / Begin	2.1	0.1	-13.6	0.0	-7.4	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_117	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-13.3	0.0	-7.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_118	SLN73 radial (ends at the ring) / Begin	1.1	-0.1	-13.3	0.0	-7.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_119	SLN73 radial (ends at the ring) / Begin	11.2	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_120	SLN73 radial (ends at the ring) / Begin	8.1	0.0	-89.8	-0.1	-47.2	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_121	SLN73 radial (ends at the ring) / Begin	8.5	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_122	SLN73 radial (ends at the ring) / Begin	7.6	0.0	-89.8	-0.1	-47.2	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_123	SLN73 radial (ends at the ring) / Begin	10.8	-0.1	-68.9	0.0	-36.3	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_124	SLN73 radial (ends at the ring) / Begin	6.7	0.1	-68.9	0.0	-36.3	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_125	SLN73 radial (ends at the ring) / Begin	6.2	0.0	-68.9	0.0	-36.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_126	SLN73 radial (ends at the ring) / Begin	4.8	-0.1	-68.9	0.0	-36.3	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_127	SLN73 radial (ends at the ring) / Begin	5.7	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_128	SLN73 radial (ends at the ring) / Begin	2.5	0.1	-20.2	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_129	SLN73 radial (ends at the ring) / Begin	1.5	0.0	-20.3	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_130	SLN73 radial (ends at the ring) / Begin	0.3	-0.1	-20.3	0.0	-10.9	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_135	SLN73 radial (ends at the ring) / Begin	7.7	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_141	SLN73 radial (ends at the ring) / Begin	2.3	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_142	SLN73 radial (ends at the ring) / Begin	1.8	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_143	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_144	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_145	SLN73 radial (ends at the ring) / Begin	10.1	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_146	SLN73 radial (ends at the ring) / Begin	8.3	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_147	SLN73 radial (ends at the ring) / Begin	9.4	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_148	SLN73 radial (ends at the ring) / Begin	9.0	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_151	SLN73 radial (ends at the ring) / Begin	6.9	-0.1	-46.4	0.0	-22.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_152	SLN73 radial (ends at the ring) / Begin	9.3	0.0	-63.5	0.0	-35.9	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_153	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.4	0.0	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_154	SLN73 radial (ends at the ring) / Begin	7.0	-0.1	-46.8	0.0	-22.5	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	3.6 < 5.0%	OK
Bolts	69.9 < 100%	OK
Welds	99.8 < 100%	OK
Buckling	Not calculated	

Plates

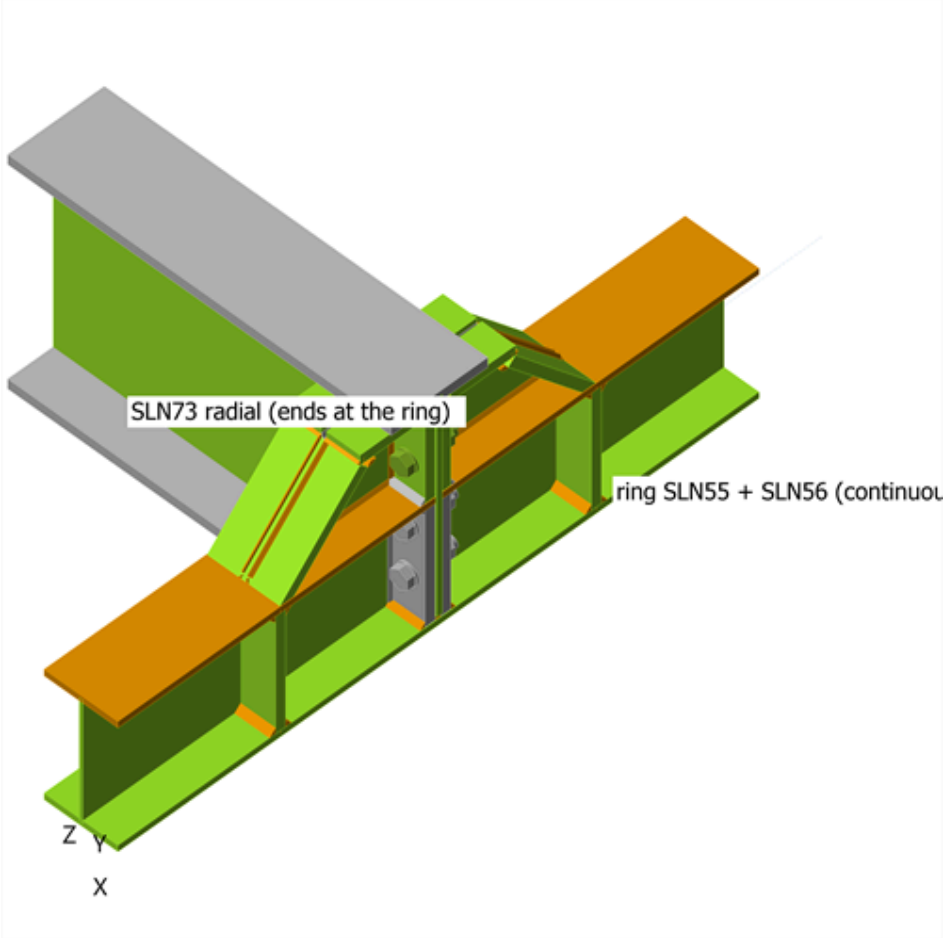
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73 radial (ends at the ring)-bfl 1	19.6	C LC_122	14.8	0.0	0.0	OK
SLN73 radial (ends at the ring)-tfl 1	19.6	C LC_119	140.0	0.0	5.8	OK
SLN73 radial (ends at the ring)-w 1	11.4	C LC_122	242.8	0.0	71.3	OK
ring SLN55 + SLN56 (continuous)-bfl 1	12.7	C LC_119	280.7	2.7	0.0	OK
ring SLN55 + SLN56 (continuous)-tfl 1	12.7	C LC_122	282.5	3.6	0.0	OK
ring SLN55 + SLN56 (continuous)-w 1	7.2	C LC_122	275.4	0.2	0.0	OK
395x145x20 fin plate upper (SLN55 side)	20.0	C LC_122	275.2	0.1	479.6	OK
395x145x20 fin plate lower outer (SLN55 side)	20.0	C LC_120	203.4	0.0	197.3	OK
395x145x20 fin plate lower inner (SLN55 side)	20.0	C LC_122	205.3	0.0	38.8	OK
395x145x20 fin plate upper (SLN56 side)	20.0	C LC_135	275.1	0.1	476.8	OK
395x145x20 fin plate lower outer (SLN56 side)	20.0	C LC_122	205.4	0.0	198.6	OK
395x145x20 fin plate lower inner (SLN56 side)	20.0	C LC_122	205.1	0.0	38.5	OK
20 ring stiffener at 315 (SLN55 side)	20.0	C LC_121	279.1	1.9	0.0	OK
10 bracket web plate (SLN55 side)	10.0	C LC_122	275.6	0.3	0.0	OK
20 bracket top plate a (SLN55 side)	20.0	C LC_119	275.0	0.0	104.0	OK
20 inclined stiffener a (SLN55 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate b (SLN55 side)	20.0	C LC_122	275.9	0.4	0.0	OK
20 inclined stiffener b (SLN55 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 ring stiffener at 315 (SLN56 side)	20.0	C LC_122	279.1	1.9	0.0	OK
10 bracket web plate (SLN56 side)	10.0	C LC_122	275.6	0.3	0.0	OK
20 bracket top plate a (SLN56 side)	20.0	C LC_120	275.9	0.4	0.0	OK
20 inclined stiffener a (SLN56 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate b (SLN56 side)	20.0	C LC_119	275.0	0.0	104.0	OK
20 inclined stiffener b (SLN56 side)	20.0	C LC_122	275.4	0.2	0.0	OK

Design data

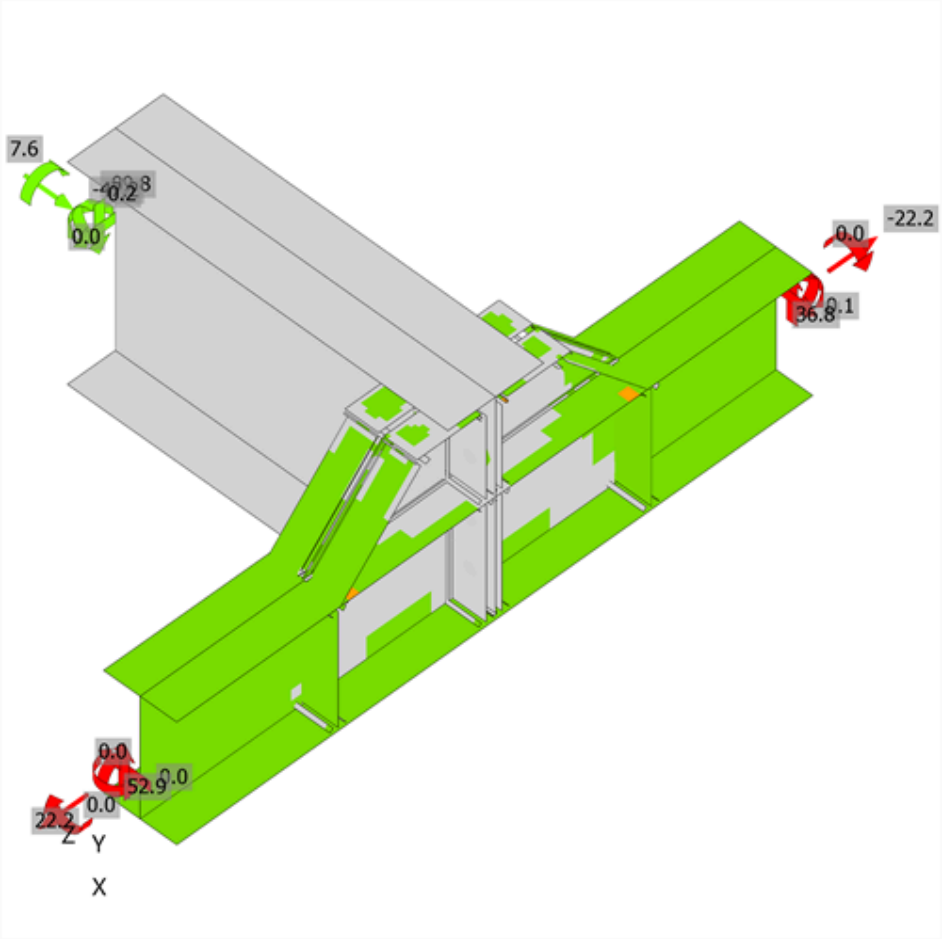
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

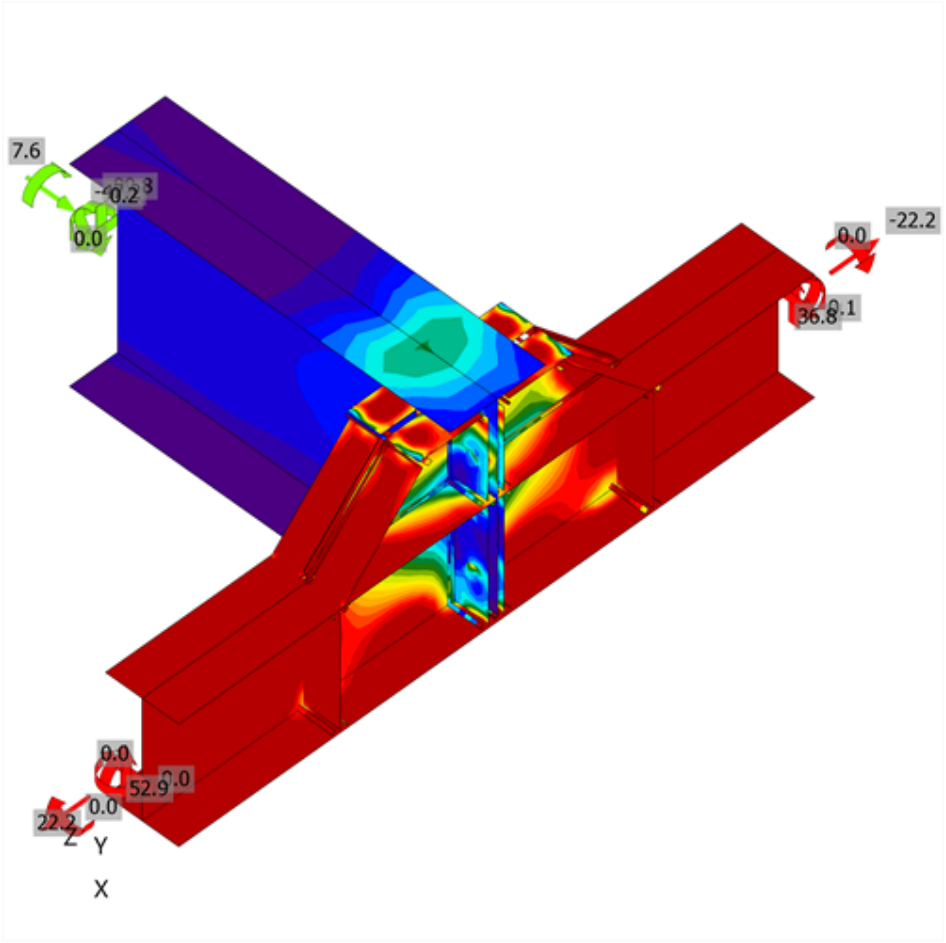
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

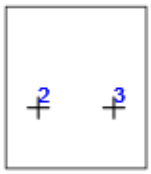
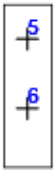
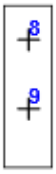


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M24 10.9 - 1	C LC_122	46.7	34.5	235.3	18.4	29.2	37.5	OK	OK
	B3	M24 10.9 - 1	C LC_122	6.7	35.2	100.7	2.6	69.9	26.8	OK	OK
	B5	M24 10.9 - 1	C LC_119	3.5	10.1	235.3	1.4	8.6	8.2	OK	OK
	B6	M24 10.9 - 1	C LC_119	1.4	32.2	235.3	0.5	27.3	23.2	OK	OK
	B8	M24 10.9 - 1	C LC_153	0.0	12.7	225.3	0.0	11.3	9.0	OK	OK
	B9	M24 10.9 - 1	C LC_119	11.7	30.8	235.3	4.6	26.2	25.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M24 10.9 - 1	254.2	492.7	141.2

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN55 side)	20 bracket top plate a (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN55 side)	20 bracket top plate b (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate a (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate b (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN55 side)	◢ 7.0 ◣	144	C LC_119	171.4	0.0	78.6	79.6	37.3	42.3	29.5	OK	OK
		◢ 7.0 ◣	144	C LC_119	146.7	0.0	-77.8	63.2	-34.2	36.3	24.0	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	69	C LC_122	255.7	0.0	6.4	-112.0	96.1	63.2	45.3	OK	OK
		◢ 7.0 ◣	69	C LC_122	396.9	0.2	-57.1	-4.2	-226.7	98.1	63.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	68	C LC_119	118.3	0.0	47.8	61.8	-8.9	29.2	22.0	OK	OK
		◢ 7.0 ◣	69	C LC_119	151.3	0.0	87.9	-71.1	-2.6	37.4	27.7	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	233	C LC_119	118.5	0.0	1.5	-1.5	68.4	29.3	20.5	OK	OK
		◢ 4.2 ◣	232	C LC_122	114.0	0.0	-29.7	8.3	63.0	28.2	19.6	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_122	384.7	0.0	37.1	73.1	208.6	95.1	49.4	OK	OK
		◢ 7.0 ◣	69	C LC_122	397.0	0.3	49.0	-33.9	-224.9	98.1	55.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_119	124.8	0.0	-21.1	-71.0	0.5	30.8	22.9	OK	OK
		◢ 7.0 ◣	69	C LC_119	151.8	0.0	-110.2	60.2	2.9	37.5	26.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	232	C LC_119	97.2	0.0	-5.9	-2.7	55.9	24.0	16.9	OK	OK
		◢ 4.2 ◣	232	C LC_122	148.9	0.0	-17.5	17.7	83.5	36.8	23.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN56 side)	◢ 7.0 ◣	144	C LC_119	148.0	0.0	-77.6	-63.7	35.1	36.6	24.1	OK	OK
		◢ 7.0 ◣	144	C LC_119	173.8	0.0	80.3	-80.4	-38.1	42.9	29.9	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_122	396.9	0.2	-56.9	4.1	226.7	98.1	63.8	OK	OK
		◢ 7.0 ◣	69	C LC_122	255.5	0.0	6.3	112.4	-95.4	63.1	45.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_119	151.5	0.0	87.5	71.4	1.9	37.4	27.8	OK	OK
		◢ 7.0 ◣	68	C LC_119	120.9	0.0	49.6	-63.0	9.1	29.9	22.5	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	232	C LC_122	113.6	0.0	-30.2	-8.9	-62.6	28.1	19.5	OK	OK
		◢ 4.2 ◣	233	C LC_119	118.5	0.0	1.6	1.3	-68.4	29.3	20.4	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_122	397.0	0.3	49.0	33.7	225.0	98.1	55.1	OK	OK
		◢ 7.0 ◣	69	C LC_122	386.3	0.0	37.8	-74.1	-209.2	95.4	49.7	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_119	152.2	0.0	-109.9	-60.7	-3.2	37.6	26.1	OK	OK
		◢ 7.0 ◣	69	C LC_119	127.1	0.0	-23.0	72.2	-0.1	31.4	23.3	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	232	C LC_122	148.2	0.0	-17.9	-18.1	-83.0	36.6	23.9	OK	OK
		◢ 4.2 ◣	232	C LC_119	97.2	0.0	-5.9	2.7	-55.9	24.0	16.9	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 8.5 ◣	147	C LC_119	396.6	0.0	74.7	145.5	-171.5	98.0	80.2	OK	OK
		◢ 8.5 ◣	146	C LC_119	285.5	0.0	-85.7	31.4	-154.1	70.6	49.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 8.5 ◣	146	C LC_122	240.3	0.0	-0.4	22.3	136.9	59.4	45.8	OK	OK
		◢ 8.5 ◣	146	C LC_153	397.3	0.4	-192.3	159.9	121.3	98.2	88.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	288	C LC_122	397.3	0.4	-27.0	-33.1	-226.5	98.2	62.1	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.4	0.5	4.0	-9.6	-229.2	98.2	69.1	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_119	403.8	4.4	132.0	-199.9	92.5	99.8	98.2	OK	OK
		◢ 7.0 ◣	67	C LC_122	402.4	3.6	146.9	207.9	-59.6	99.4	91.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_119	402.4	3.6	-113.7	196.4	105.2	99.4	93.6	OK	OK
		◢ 7.0 ◣	67	C LC_121	402.7	3.8	-137.3	-208.8	-64.6	99.5	88.2	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 8.5 ◣	147	C LC_119	396.7	0.0	74.8	145.1	171.8	98.0	77.2	OK	OK
		◢ 8.5 ◣	146	C LC_119	286.4	0.0	-85.8	31.5	154.6	70.8	50.5	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 8.5 ◣	146	C LC_122	240.1	0.0	-0.5	22.1	-136.9	59.3	46.7	OK	OK
		◢ 8.5 ◣	146	C LC_153	397.3	0.4	-192.4	160.0	-121.1	98.2	88.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	288	C LC_122	397.4	0.5	4.1	9.3	229.3	98.2	69.1	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.3	0.4	-26.7	32.7	226.5	98.2	62.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_119	402.4	3.6	-114.4	196.6	-104.6	99.4	94.5	OK	OK
		◢ 7.0 ◣	67	C LC_122	402.7	3.8	-137.2	-208.9	64.5	99.5	87.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_119	403.8	4.5	132.7	-200.1	-92.0	99.8	97.4	OK	OK
		◢ 7.0 ◣	67	C LC_122	402.4	3.6	147.5	207.8	59.4	99.4	92.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
395x145x20 fin plate upper (SLN55 side)	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	167	C LC_119	399.4	1.7	-302.0	110.2	103.1	98.7	77.6	OK	OK
		◢ 4.2 ◣	167	C LC_119	400.3	2.3	134.9	187.7	110.1	98.9	80.8	OK	OK
395x145x20 fin plate upper (SLN56 side)	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	167	C LC_119	400.3	2.3	135.4	-187.8	-109.8	98.9	80.5	OK	OK
		◢ 4.2 ◣	167	C LC_119	399.5	1.8	-301.2	-110.8	-103.4	98.7	77.7	OK	OK
20 bracket top plate a (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.5	0.5	-48.7	2.4	227.7	98.2	79.0	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.7	0.0	-55.3	16.9	226.1	98.0	67.8	OK	OK
20 bracket top plate b (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.2	0.4	7.9	-31.9	227.1	98.2	77.4	OK	OK
		◢ 7.0 ◣	67	C LC_122	384.2	0.0	-68.0	31.7	216.0	94.9	67.5	OK	OK
20 bracket top plate a (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.2	0.4	7.6	-31.8	-227.1	98.2	74.6	OK	OK
		◢ 7.0 ◣	67	C LC_122	384.2	0.0	-67.1	30.8	-216.2	94.9	62.2	OK	OK
20 bracket top plate b (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.5	0.5	-48.8	2.2	-227.7	98.2	77.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.7	0.0	-56.4	17.4	-226.0	98.0	69.9	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.5	1.2	-261.5	-147.1	-92.1	98.5	92.4	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.6	1.2	-81.4	189.5	121.9	98.5	78.2	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.9	1.4	129.6	122.3	-180.2	98.6	93.5	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.2	1.0	164.8	-174.8	115.1	98.4	71.2	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.9	1.4	130.0	122.5	180.0	98.6	93.3	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.2	1.0	164.7	-174.8	-115.2	98.4	75.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.5	1.2	-261.2	-147.2	92.3	98.5	91.1	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.6	1.2	-81.7	189.3	-122.0	98.5	76.8	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate a (SLN55 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-199.1	-188.1	-62.6	98.1	71.0	OK	OK
		◢ 4.2 ◣	67	C LC_122	397.0	0.2	231.5	-164.0	-88.1	98.1	78.8	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate b (SLN55 side)	◢ 4.2 ◣	67	C LC_122	402.8	3.8	-233.8	189.3	4.0	99.5	96.8	OK	OK
		◢ 4.2 ◣	67	C LC_122	402.8	3.8	-224.2	188.9	-40.6	99.5	95.2	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate a (SLN56 side)	◢ 4.2 ◣	67	C LC_120	402.5	3.6	-234.3	188.9	-3.0	99.4	96.7	OK	OK
		◢ 4.2 ◣	67	C LC_120	402.5	3.6	-224.8	188.3	41.4	99.5	93.0	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate b (SLN56 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-200.6	-187.3	63.2	98.1	71.8	OK	OK
		◢ 4.2 ◣	67	C LC_122	397.0	0.2	232.1	-163.6	88.4	98.1	76.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{C}	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5